

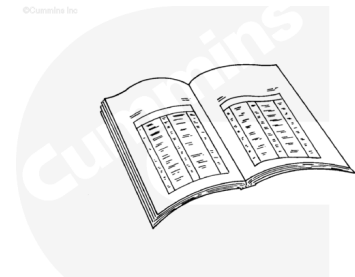
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

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ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and caused reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative battery cable last.

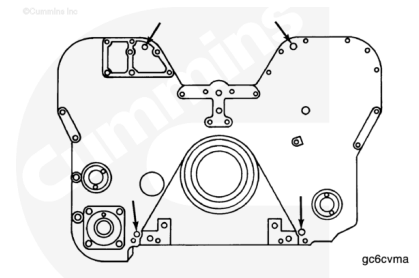
- Disconnect the batteries. Refer to Procedure 013-009 in Section 1. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Drain the engine oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>)
- Disconnect the air/fuel supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/102/102-012-022.html>)
- Remove the fan hub idler assembly. Refer to Procedure 008-029 in Section 8 (</qs3/pubsys2/xml/en/procedures/28/28-008-029-tr.html>).
- Remove the water pump drive. Refer to Procedure 008-062 in Section 8 (</qs3/pubsys2/xml/en/procedures/28/28-008-062-tr.html>).
- Remove the fuel pump/air compressor drive. Refer to Procedure 005-016 in Section 5 (</qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html>).
- Remove the hydraulic pump drive. Refer to Procedure 009-016 in Section 9 (</qs3/pubsys2/xml/en/procedures/28/28-009-016-tr.html>).
- Remove the vibration damper. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-052-tr.html>)
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/28/28-013-005-tr.html>)
- Remove the alternator bracket. Refer to Procedure 013-003 in Section 13.

(/qs3/pubsys2/xml/en/procedures/28/28-013-003-tr.html)

- Remove the alternator. Refer to Procedure 013-001 in Section 13. (/qs3/pubsys2/xml/en/procedures/28/28-013-001-tr.html)
- Remove the LTA water pump. Refer to Procedure 008-109 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-109-tr.html)
- Remove the oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)

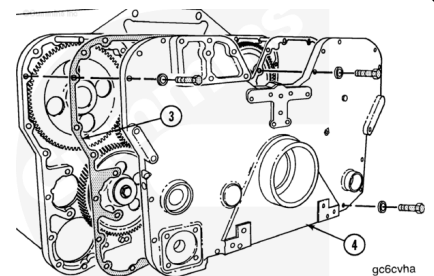
Remove

There are four threaded holes in the cover. Two of them are near the top and two of them are at the bottom. They can be used to push the gear cover from the engine. The holes have [7/16-14] threads.



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Note : One capscrew is located to the right bank rear of the front gear housing.

Remove all the capscrews.

Remove the gear cover and gasket.

Discard the gasket.

The dowel pins do **not** have to be removed unless they are damaged.

Disassemble

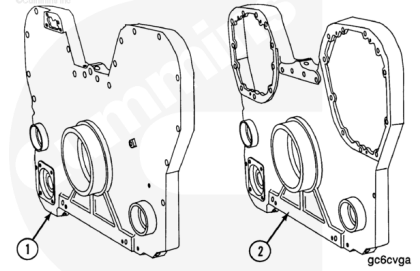
There are two different front gear covers used on K38 and K50 engines.

The solid cover (1) is on most engines.

The other cover (2) is machined to mount sea (raw) water pumps for marine engines. If the sea water pumps are **not** installed, the cover (2) has cover plates. The covers are interchangeable.

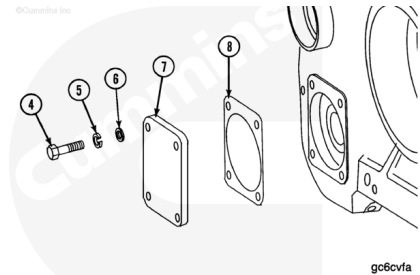
The cover (2) allows the camshaft or camshaft gear to be removed, without removing the gear cover.

The procedures and specifications in this procedure apply to both types of covers. **Only** the solid cover (1) is illustrated.



⚠ CAUTION ⚠

Engines not equipped with hydraulic pumps must contain a flange plug. This part has a plug welded to the cover (7). The plug blocks the oil drilling in the bushing. If the drilling is not blocked, the engine oil pressure will be less than required specifications.



Remove the parts.

4 Capscrews

5 Lock washers

6 Flat washers

7 Flange cover

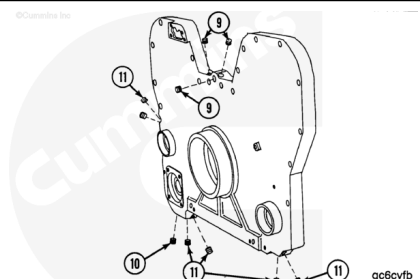
8 Gasket

Remove the three pipe plugs (9), the one pipe plug (10), and the six plugs (11).

Note : Newer covers will contain AVSEAL™ instead of pipe plugs (11) in the lower oil drillings. Use the following procedure for instructions to remove the AVSEAL™. Modify the oil drillings with AVSEAL™ Remove/Install Kit, Part Number 3823154. Refer to Procedure 001-026 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)

Note : Newer covers do not contain the pipe plug (10).

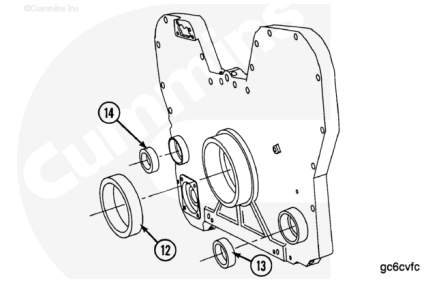


Use a mandrel or a drift. Use a mallet.

Remove the oil seals.

1. Front crankshaft seal
2. Accessory drive seal
3. Water pump drive seal

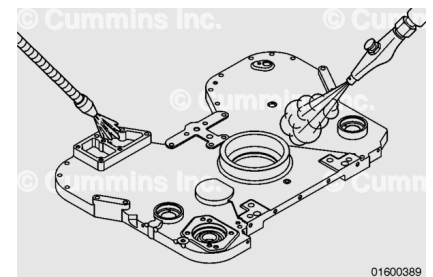
Note : The water pump and accessory drive seals are the same part number. Discard the seals.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

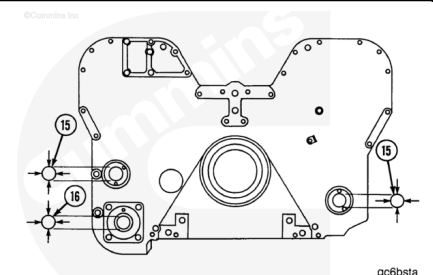
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam. Clean the parts.

Dry with compressed air.

Note : There is no gear cover bushing for the water pump on engines equipped with dual impeller water pumps.

Measure the inside diameter of the bushing.



Replace the bushing if it is **not** within specifications. Refer to Procedure 001-066 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-066-tr.html)

Bushing Inside Diameter			
	mm		in
(15) Accessory Drive and Water Pump Drive:	39.75	MIN	1.565
	39.790	MAX	1.571
(16) Hydraulic Pump Drive:	38.13	MIN	1.501
	38.25	MAX	1.506

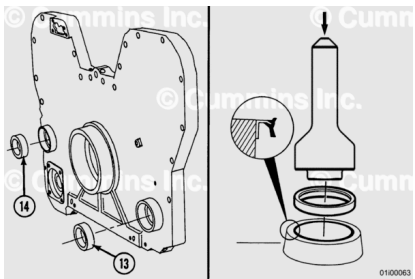
Check the trunnion sleeve for damage. The sleeve **must** be replaced if it is worn or cracked. Refer to Procedure 001-030 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-030-tr.html)

Assemble

Use a seal installer, Part Number 5573467. Install the accessory drive seal (13), and the water pump seal (14).

Push the seals into the cover until the service tool flange stops against the gear cover.

Note : Do not install the crankshaft seal until the front cover is installed on the engine.



Install the pipe plugs (9), (10), and (11).

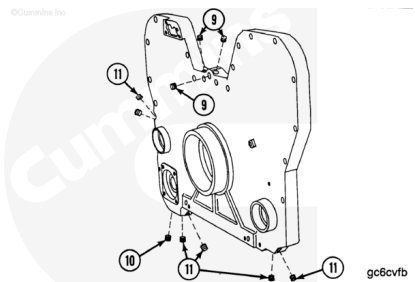
Torque Value:

[0.125 in]

1. 20 n•m [177 in-lb]

Torque Value:

[0.250 in]



1. 27 n•m [239 in-lb]

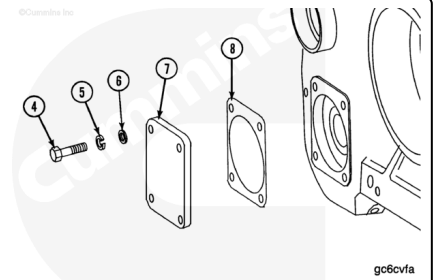
Torque Value:

[0.375 in]

1. 34 n•m [25 ft-lb]

Install the flange cover (7) and gasket (8). Install the plain washers (6), lock washers (5), and capscrews (4).

Torque Value: 61 n•m [45 ft-lb]



Install

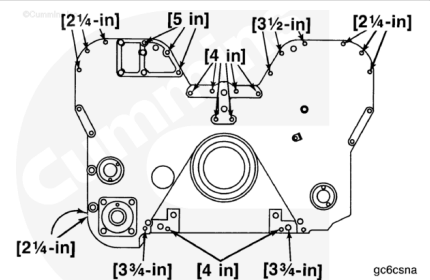
Use the following procedure for information on plugs for the front gear cover. Refer to Procedure 001-026 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)

Note : The housing shown in this procedure is typical. If the engine being serviced has sea water pumps, the housing is different. Some of the capscrews shown are different for the sea water pump supports and a front power take-off housing.

This illustration shows the capscrew lengths required for a typical cover. All of the capscrews are [7/16-14 UNC] The [5 in] capscrews are **only** installed when a fan belt idler is **not** installed.

Note : The [2.250 in] capscrew in the lower left of the illustration is installed from the gear housing side.

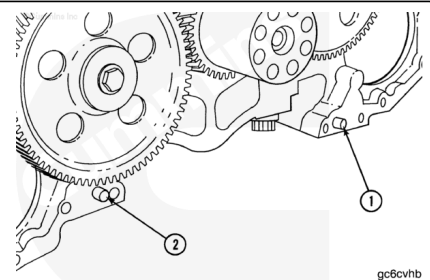
Note : If the engine contains a front power take-off housing, do **not** install the [4 in] capscrews at this time.



Check the protrusion of the diamond dowel pin (1), and master dowel pin (2) from the front gear housing. Refer to Procedure 001-026 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)

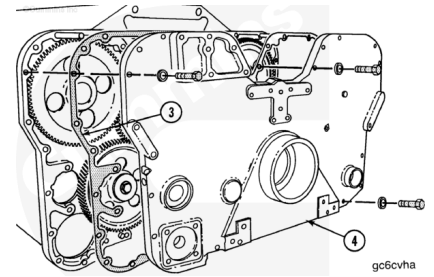
Check to be sure the mating surfaces of the gear housing are clean, dry, and free of nicks and burrs.



The use of guide bolts will aid with the assembly process.

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Install the gasket (3) on the gear housing.

Install the cover (4), lock washers, and capscrews.

Use a mallet. Tap the cover on the dowel pins until the cover is touching the housing.

Torque Value: 60 n•m [44 ft-lb]

Finishing Steps

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Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-

027-tr.html)

- Install the LTA water pump. Refer to Procedure 008-109 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-109-tr.html)
- Install the alternator. Refer to Procedure 013-001 in Section 13. (/qs3/pubsys2/xml/en/procedures/28/28-013-001-tr.html)
- Install the alternator bracket. Refer to Procedure 013-003 in Section 13. (/qs3/pubsys2/xml/en/procedures/28/28-013-003-tr.html)
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13. (/qs3/pubsys2/xml/en/procedures/28/28-013-005-tr.html)
- Install the vibration damper. Refer to Procedure 001-052 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-052-tr.html)
- Install the hydraulic pump drive. Refer to Procedure 009-016 in Section 9. (/qs3/pubsys2/xml/en/procedures/28/28-009-016-tr.html)
- Install the water pump drive. Refer to Procedure 008-062 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-062-tr.html)
- Install the fuel pump/air compressor drive. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html)
- Install the fan hub idler assembly. Refer to Procedure 008-029 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-029-tr.html)
- Connect the air/fuel supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (/qs3/pubsys2/xml/en/procedures/102/102-012-022.html)
- Fill the engine with oil. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 1. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)

- Operate the engine and check for leaks.

Last Modified: 26-Sep-2022
